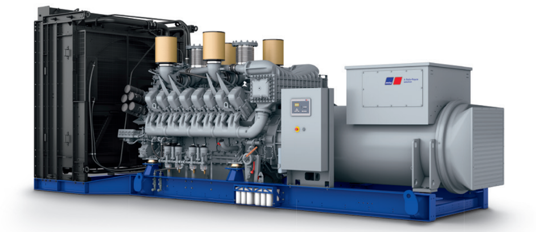
Diesel Generator Set

mtu 16V4000 DS2250

380V – 11 kV/50 Hz/prime power for stationary emergency/ NOx emission optimized/16V4000G14F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits
- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support
- Global product support offered

Standards
- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating
- System ratings: 2050 kVA - 2160 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)
- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available
- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions
- NOx emission optimized

Certifications
- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

Application data [1)]

Engine

Manufacturer	*mtu*
Model	16V4000G14F
Type	4-cycle
Arrangement	16V
Displacement: l	76.3
Bore: mm	170
Stroke: mm	210
Compression ratio	16.4
Rated speed: rpm	1500
Engine governor	ECU 9
Max power: kWm	1798
Air cleaner	dry

Fuel system

Maximum fuel lift: m	5
Total fuel flow: l/min	20

Fuel consumption [2)]	l/hr	g/kwh
At 100% of power rating:	472.2	218
At 75% of power rating:	346.1	213
At 50% of power rating:	234	216

Liquid capacity (lubrication)

Total oil system capacity: l	300
Engine jacket water capacity: l	175
Intercooler coolant capacity: l	50

Combustion air requirements

Combustion air volume: m³/s	3.2
Max. air intake restriction: mbar	50

Cooling/radiator system

Coolant flow rate (HT circuit): m³/hr	68.5
Coolant flow rate (LT circuit): m³/hr	30
Heat rejection to coolant: kW	790
Heat radiated to charge air cooling: kW	460
Heat radiated to ambient: kW	90
Fan power for electr. radiator (40°C): kW	70

Exhaust system

Exhaust gas temp. (after turbocharger): °C	480
Exhaust gas volume: m³/s	7.4
Maximum allowable back pressure: mbar	85
Minimum allowable back pressure: mbar	30

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NOx emission optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S7 (Low voltage Leroy Somer standard)	380 V	1728	2160	3282	1656	2070	3145
	400 V	1728	2160	3118	1656	2070	2988
	415 V	1728	2160	3005	1656	2070	2880
Marathon 744RSL7092 (Low voltage Marathon)	380 V	1704	2130	3236	1640	2050	3115
	400 V	1704	2130	3074	1640	2050	2959
	415 V	1696	2120	2949	1640	2050	2852
Marathon 1020FDL7093 (Low voltage Marathon oversized)	380 V	1704	2130	3236	1640	2050	3115
	400 V	1704	2130	3074	1640	2050	2959
	415 V	1696	2120	2949	1640	2050	2852
Marathon 1020FDH7097 (Medium volt. marathon)	11 kV	1712	2140	112	1640	2050	108
Leroy Somer LSA53.2 XL9 (Med. volt. Leroy Somer)	11 kV	1728	2160	113	1656	2070	109

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.